

113B Data Science and Python Programming

Homework Assignment #0

Due: 2025/3/18 12:00:00

Since this is an advanced course for the student who has already passed the introductory course of basic programming with Python, in this homework, you are challenged to write a program that returns the answer to the corresponding question.

Problem #1: Warm up

Please write a program that answers the following questions:

1. Why are you here?
2. Why do you want to take this course?
3. What do you expect to learn in this course?
4. Anything you would like to say.

Here is the sample code:

```
class hw0:
    def __init__(self, student_name="Bob Lin",
                  student_id="0123456789",
                  department="Department of Photonics"):
        self.name = student_name
        self.id = student_id
        self.dpt = department

    def __str__(self):
        return "Student: {}, {}, {}".format(self.name, self.id,
                                              self.dpt)

    def ???(self):
        剩下你自己想
```

Template for your main script:

```
import argparse
from ans import hw0

if __name__ == "__main__":
    parser = argparse.ArgumentParser(description="Homework #0")
```

```

    parser.add_argument("--num", default=0, type=int, help="Return which
question? (default: 0)")
    opt = parser.parse_args()

    myAns = hw0()
    print("="*100)
    print(myAns)
    print("="*100)

```

剩下你自己想

Problem #2: Summarize your result

Please printscreen and summarize the results of every question in your report. For example,

```

(nycudops) 0: \113B\HW0>python main_hw0.py
=====
Student: 000的小三, 0123456789, Department of Photonics
=====
Question 1: Why are you here?
Response:
    我偶然在二餐看到了000
Question 2: Why do you want to take this course?
Response:
    因为我深深地被000吸引
Question 3: What do you expect to learn in this course?
Response:
    講幹話的藝術
Question 4: Anything you would like to say.
Response:
    蕭博勁救救我

```

Please accomplish this homework with an organized code (e.g., with main script and function script). Here is a template for your code structure:

```

113B_hw0_0123456789
├─ ans.py          # Solution of homework #0
├─ report.pdf      # Report of homework #0
└─ main_hw0.py     # Main script of homework #0

```

You don't need to follow this structure, just keep your main script clean.

FRIENDLY REMINDER

Response in English is not necessary.

Feel free to choose any language you want!

Your report should be a .pdf file with filename **report.pdf.**

Hand in procedure:

As we had mentioned in the lecture, you should list all your collaborators in your programs (e.g., signature). **Your signature should be attached in your main script.** Here is the template of signature:

```
""  
Created on Wed Aug 7 01:23:45 2024  
  
@author: Xi Winnie, student ID  
  
@collaborators: William Lai, his student ID  
""
```

Please save your code as a “.zip”, “.7z”, or “.rar” file, where the file name should follow this format:

113B_hw0_ID.zip

For example,

113B_hw0_0123456789.zip

Please be aware. **We are not going to accept any homework file with wrong file name or without signature.** Please double check the content of your file.

Once you have accomplished your works, you can upload your homework to the “E3@NYCU” system. There will be a section for uploading your homework.